

InStore Optima — Development Changes Summary

Branch: dev1

Period: May 2026

Project: Full-stack Inventory Management System (React + [ASP.NET](#) Core 8 + SQL Server)

Overview

This document summarises all the features, fixes, and improvements made to the **InStore Optima** client application during this development session. Changes span across frontend (React/Vite) and backend ([ASP.NET](#) Core 8 with Serilog).

1. Form Validations — All Pages

What changed:

Added complete inline field-level validations and toast error notifications across every page in the application.

Pages covered:

Login , Register , Products , Suppliers , Stock , Orders , OrderItems , Invoices , Receipts , Payments , PurchaseOrders , Replenishment , StockMovement , Users

Behaviour:

- Each form field shows a red error message directly beneath it when invalid
- A toast notification also appears at the top with the first error message
- Fields clear their error state as soon as the user starts correcting them (live feedback)

2. Live Validation — Login & Register Pages

What changed:

Login and Register pages now validate fields in real-time as the user types, not just on form submit.

Behaviour:

- Email format, password length, name requirements are checked on every keystroke
- Errors appear and disappear instantly without needing to click the submit button

3. Global Exception Handling — Frontend

What changed:

Added two layers of frontend error handling:

a) React Error Boundary (`ErrorBoundary.jsx`)

- Wraps the entire application in `App.jsx`
- Catches any unexpected JavaScript runtime errors in any component
- Shows a friendly "Something went wrong" fallback UI instead of a blank/crashed screen
- Logs the error to the console for debugging

b) API Error Parser (`parseApiError` in `validators.js`)

- Centralised function used across all pages to extract meaningful messages from API errors
- Detects when the **backend server is not running** (network errors, 502/503 status) and shows:
"Backend is not live. Please start the backend server and try again."
- Handles timeouts, validation errors, and all other HTTP error responses gracefully

4. Show/Hide Password Toggle

What changed:

Login and Register pages now have a show/hide password toggle button (eye icon) on password fields.

Behaviour:

- Click the eye icon to reveal the password as plain text
- Click again to hide it
- Works on both the Password and Confirm Password fields on Register

5. Serilog Logging — Backend

What changed:

Integrated **Serilog** into the [ASP.NET](#) Core 8 backend for structured application logging.

Packages added:

- Serilog.AspNetCore 10.0.0
- Serilog.Sinks.Console 6.1.1
- Serilog.Sinks.File 7.0.0

Configuration:

- Logs are written to the **console** (for development visibility)
- Logs are also written to **rolling daily files** at `Logs/log-YYYYMMDD.txt`
- All unhandled exceptions pass through `GlobalExceptionHandlerMiddleware` which logs them via `_logger.LogError(...)`

6. Global Exception Handling Middleware — Backend

What changed:

Added `GlobalExceptionHandlerMiddleware.cs` to the [ASP.NET](#) Core pipeline.

Behaviour:

- Catches all unhandled exceptions before they reach the client
- Returns structured JSON error responses with appropriate HTTP status codes:
 - 404 — Resource not found
 - 400 — Validation errors
 - 401 — Unauthorised access
 - 409 — Conflict (e.g. duplicate entries)
 - 500 — Internal server errors

- All errors are logged via Serilog

7. IST Timezone Fix — Date/Time Display

What changed:

All dates and timestamps across the application now correctly display in **Indian Standard Time (IST, UTC+5:30)**.

Root cause:

The backend returned UTC timestamps without the `Z` suffix (e.g. `"2026-05-10T14:30:00"` instead of `"2026-05-10T14:30:00Z"`), causing browsers to misinterpret them as local time.

Fix:

Added a `toUTC()` helper inside `validators.js` that appends `Z` if missing before parsing, then formats with `timeZone: 'Asia/Kolkata'` .

Functions added to `validators.js` :

- `fmtDate(val)` — formats a date as `DD/MM/YYYY` in IST
- `fmtDateTime(val)` — formats a date-time as `DD/MM/YYYY, HH:MM:SS AM/PM` in IST

Pages using these:

`AuditLogs` , `Stock` , `Orders` , `Invoices` , `Receipts` , `Payments` , `PurchaseOrders` , `Replenishment` , `StockMovement` , `Users`

8. Toast Notifications — Always Visible Above Modals

What changed:

Toast notifications were being hidden behind open modals (z-index conflict).

Fix:

`useToast.jsx` now renders toasts via `ReactDOM.createPortal` directly into `document.body` with `z-index: 999999` , ensuring they always appear on top of any modal or overlay.

9. Out-of-Stock Filter — Order Items Page

What changed:

The product dropdown in the **Add Item** form on the Order Items page now only shows products that are **currently in stock**.

Behaviour:

- Products with `currentStock === 0` are hidden from the dropdown
- Products without a stock record are also hidden
- The dropdown shows available stock count: "Product Name – ₹price (Stock: 5)"
- Stock data is **refreshed from the API every time** the "Add Item" modal is opened, so it always reflects the latest stock levels

10. Auto-Replenishment on Low Stock

What changed:

The system now automatically creates a replenishment order when stock falls critically low.

Trigger condition:

`currentStock >= 0 AND currentStock <= floor(minStock / 2)`

Triggered from two places:

a) Stock Page (manual stock edit)

- After updating or creating a stock entry, the system checks the new stock level
- If it crosses the threshold, a replenishment order is auto-created for `minStock × 2` units

b) Order Items Page (after placing an order)

- After a new order item is saved, the backend deducts stock
- The frontend then fetches fresh stock and checks the threshold
- If triggered, a replenishment order is auto-created and a **warning toast** is shown:
"Item added! Auto-replenishment created – stock critically low ($23 \leq 25$)."

11. Purchase Order Date Validation

What changed:

The **Create Purchase Order** form now enforces that the delivery date cannot be in the past.

Behaviour:

- The date picker calendar blocks past dates (via `min` attribute set to today's date)
- If a past date is typed manually, a validation error appears:
"Delivery date cannot be in the past."

12. Filter Used Replenishments — Purchase Orders

What changed:

When creating a Purchase Order, the **Replenishment Order** dropdown now only shows replenishments that:

1. Have status `Approved`
2. Have **not already been linked** to an existing Purchase Order

This prevents the same replenishment from being referenced by multiple Purchase Orders.

13. Backend Not Live — User-Friendly Error Messages

What changed:

When the backend API server is not running, instead of showing a cryptic network error, the app shows:

"Backend is not live. Please start the backend server and try again."

How it works:

- `axiosClient.js` intercepts `ERR_NETWORK` errors and connection refused responses
- `parseApiError()` in `validators.js` also checks for HTTP 502/503 status codes
- Both show the above message to the user via toast

Tech Stack Reference

Layer	Technology
Frontend	React 18, Vite, Bootstrap Icons
Backend	ASP.NET Core 8, Entity Framework Core
Database	SQL Server (LocalDB)
Auth	JWT Bearer Tokens + BCrypt
Logging	Serilog (Console + File sinks)
HTTP Client	Axios
State	React Context (Auth, Theme)

File Reference — Key Files Changed

File	Changes
src/utils/validators.js	Added fmtDate , fmtDateTime , parseApiError , validateField , validateForm
src/api/axiosClient.js	Network error handling, timeout handling, 401 redirect
src/hooks/useToast.jsx	Portal-based rendering for modal z-index fix
src/components/shared/ErrorBoundary.jsx	New — React class error boundary
src/pages/OrderItems.jsx	Out-of-stock filter, auto-replenishment trigger
src/pages/Stock.jsx	Auto-replenishment trigger on stock edit
src/pages/PurchaseOrders.jsx	Date validation, used-replenishment filter
src/pages/Login.jsx	Live validation, show/hide password
src/pages/Register.jsx	Live validation, show/hide password

File	Changes
All other pages	Inline field errors + toast on validation failure + IST date formatting
Program.cs (backend)	Serilog setup
GlobalExceptionHandlerMiddleware.cs (backend)	Global error handling + logging